

PROBLEM STATEMENT

problem stems from the severe inefficiency and lack of privacy inherent in traditional campus Xerox shops, where students routinely face 20+ minute wait times due to human-operated bottlenecks and peak-hour congestion. This manual workflow is plagued by logistical friction, such as the need for physical USB drives or insecure WhatsApp/Email file transfers, and operational rigidities like limited shop hours and fixed pricing that doesn't distinguish between high-cost color and low-cost black-and-white prints. Furthermore, traditional shops offer zero data security, exposing sensitive student documents to shopkeepers and public-access computers. By failing to provide a 24/7, autonomous, and UPI-integrated solution, the current system creates a "productivity gap" where students lose valuable academic time to a mundane administrative task

SOLUTION

The solution developed by NK Design House is a fully integrated, "human-less" IoT kiosk designed to convert the traditional, friction-heavy printing process into a seamless, high-speed digital utility. By housing a high-performance laser engine within a secure, industrial-grade Lavender chassis, we eliminate the need for shopkeepers and physical storefronts entirely. The core of the solution lies in its decentralized architecture: a Raspberry Pi 5 acts as the localized brain, using a secure Cloudflare Zero Trust tunnel to host a private web portal. Students simply scan a unique QR code on the machine to upload their documents via their smartphones, bypassing the need for USB drives or public access computers.

To solve the system features a Dynamic Logic Engine that automatically calculates page counts and generates a real-time UPI payment gateway link. Only upon confirmed payment does the file move from the Pi's RAM disk storage (ensuring 100% data privacy) to the print spooler. This automation allows for 24/7 availability, significantly lower operating costs by removing labor expenses, and a dramatic reduction in wait times from 20 minutes to under 60 seconds. By separating Color and B&W billing through software, we provide a fair, transparent, and high-tech printing experience that is perfectly suited for high-density environments like the SIT Campus.

IDEATION

A student scans a QR code at the kiosk to open a web portal where they upload their file. The system validates the file and prompts for print specifications: number of copies, duplexing, and color mode. The color selection includes a unique "Color X" feature, allowing users to specify specific pages for color printing while the rest remain B/W. After confirming the calculated cost (₹2 for B/W, ₹10 for Color), the user pays via GPay.

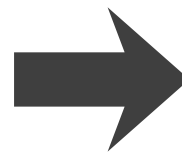
Once payment is confirmed, the file is securely transmitted to a Raspberry Pi via Cloudflare Tunnels. The student receives a unique Token QR code. To print, the student scans this Token QR at the kiosk's barcode scanner. After the student confirms the file on the kiosk monitor, the Raspberry Pi executes the print command based on the JSON-stored specifications.

OVERVIEW

This document outlines the technical workflow for engineers to implement and maintain the smart printing kiosk system. The system enables students to upload files, configure print specifications, pay digitally, and securely print documents via a Raspberry Pi.

SOFT SIDE - FILE UPLOAD & VALIDATION

- Action: Student scans QR code at kiosk.
- System Response: Opens web portal.
- Process:
 - Student uploads file.
 - System validates file format and integrity.



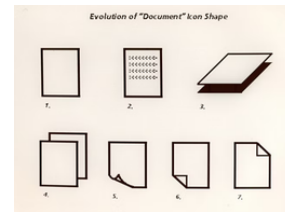
SOFT SIDE - PRINT SPECIFICATIONS

Options Provided:

- Number of copies.
- Duplexing (single/double-sided).

Color mode:

- B/W (₹2 per page).
- Color (₹10 per page).
- Color X: Specific pages selected for color printing, rest remain B/W.
- System Task: Calculate total cost based on selections.



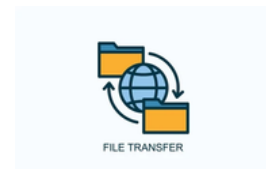
SOFT SIDE - PAYMENT

- Method: GPay integration.
- Process:
 - Student confirms cost.
 - Payment processed via GPay.
 - System verifies payment success.



SOFT SIDE - SECURE FILE TRANSMISSION

- Action: After payment confirmation.
- Process:
- File transmitted securely to Raspberry Pi using Cloudflare Tunnels.
 - JSON file created with print specifications.



SOFT SIDE - TOKEN GENERATION

- System Response: Generates unique Token QR code.
- Student Action: Receives Token QR code.

SOFT SIDE - PRINTING AT KIOSK

Process:

- Student scans Token QR at kiosk barcode scanner.
- Kiosk monitor displays file for confirmation.
- Student confirms.
- Raspberry Pi executes print command using JSON specifications.



NK-001



SECURITY & RELIABILITY NOTES

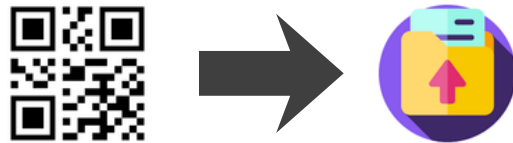
- File Security: Cloudflare Tunnels ensure encrypted transmission.
- Authentication: Token QR ensures only authorized student can print.
- Error Handling:
 - Invalid QR → prompt re-scan.
 - Payment failure → retry option.
 - Print error → log and notify support.

ENGINEER RESPONSIBILITIES

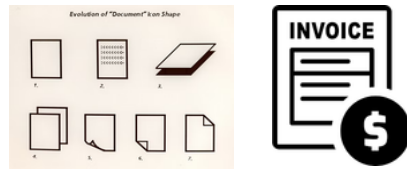
1. Maintain QR code generation and validation logic.
2. Ensure GPay integration is reliable and secure.
3. Configure Raspberry Pi to parse JSON and execute print commands.
4. Monitor Cloudflare Tunnel connections.
5. Implement logging and error recovery mechanisms.

WORKFLOW DIAGRAM

QR SCAN → WEB PORTAL → FILE UPLOAD → VALIDATION



PRINT SPECS → COST CALCULATION → PAYMENT VIA GPAY



PAYMENT SUCCESS → SECURE TRANSMISSION → JSON SPECS



TOKEN QR GENERATED → STUDENT SCANS AT KIOSK



NK-001

FILE CONFIRMATION → RASPBERRY PI EXECUTES PRINT



CONCLUSION

This workflow ensures a secure, user-friendly, and efficient printing process for students while providing engineers with clear responsibilities for system reliability and maintenance.